TELKOM UNIVERSITY • SCHOOL OF COMPUTING
COURSE: CAK4WBB3 – COMPUTER VISION | ODD SEMESTER 2024/2025

TASK 1: OBSERVATION OF CNN ARCHITECTURES IN OBJECT CLASSIFICATION

Performance, Transfer Learning Dynamics, and Computational Efficiency of MobileNetV2 for Multiclass Recyclable Garbage Classification

An Empirical Study of Depthwise Separable Convolutions & Linear Inverted Bottlenecks under Resource Constraints

EXECUTIVE SUMMARY & ABSTRACT:
Automated sorting of recyclable municipal solid waste represents a vital environmental application of modern computer vision. In this study, we implement, optimize, and rigorously analyze **MobileNetV2**—a lightweight convolutional neural network engineered for edge and mobile deployment—on a 5-class garbage classification dataset comprising 2,390 images (*cardboard, glass, metal, paper, plastic*). Utilizing a two-phase transfer learning framework (frozen feature extraction followed by selective fine-tuning of upper inverted residual stages), we evaluate classification accuracy, macro/weighted F1-scores, class-specific confusion dynamics, parameter footprint, and CPU latency. Our empirical findings demonstrate that MobileNetV2 achieves a strong test accuracy and macro F1-score with merely **2.23M parameters** and a compact **8.65 MB** model footprint, executing inference at ~25 ms per image on general-purpose CPU. These results confirm the practical viability of MobileNetV2 for automated sorting stations and resource-constrained edge recycling facilities.

Parameter / Attribute	Specification / Experimental Setup
Target Architecture	MobileNetV2 (Sandler et al., CVPR 2018) with Inverted Residuals
Pretraining Weights	ImageNet-1K (MobileNet_V2_Weights.IMAGENET1K_V1)
Dataset & Classes	Garbage Classification Dataset (5 classes: Cardboard, Glass, Metal, Paper, Plastic)
Total Dataset Size	2,390 validated JPEG images (100% data integrity verified via PIL)
Data Partitioning	Stratified 70% Train (1,672) / 15% Val (359) / 15% Test (359) • Seed=42
Training Paradigm	Two-Phase: Phase 1 (Frozen Backbone, lr=1e-3) + Phase 2 (Fine-Tune features[14:], lr=1e-4)
Loss & Optimization	Class-Weighted CrossEntropyLoss + Adam (weight decay=1e-4) + ReduceLROnPlateau
Execution Environment	Anti-Gravity IDE • Python 3.13.2 • PyTorch 2.14.1 (CPU execution)
Primary Performance Metrics	Accuracy: **92.76%** • Macro F1: **92.75%** • Latency: **15.47 ms/img**

Author / Student: Computer Vision Research Team (Branch: MobileNetV2)
Submission Date: October 2026 • Telkom University

1. Introduction and Problem Formulation

Efficient waste management and automated material recovery are critical components of sustainable urban infrastructure. Manual sorting of post-consumer recyclables is labor-intensive, hazardous, and error-prone. While convolutional neural networks (CNNs) have achieved remarkable classification capabilities, standard models such as VGG-16 (138M parameters) or ResNet-50 (25.6M parameters) demand substantial memory and compute budgets. This study evaluates **MobileNetV2**—a lightweight architecture optimized for mobile and embedded inference—on the multiclass classification of recyclable materials under strict CPU resource constraints.

2. Dataset Exploration, Integrity, and Preprocessing

The experimental dataset consists of five primary recyclable categories: *Cardboard, Glass, Metal, Paper*, and *Plastic*. A sixth category present in the raw directory (*Trash*, 137 images) was pruned to eliminate severe class imbalance (which exceeded 4.3×) and align with university benchmark splits. Every image underwent programmatic verification via Pillow to confirm complete decodability and absence of header truncation, yielding **2,390 valid JPEG images**.

Category	Images	Train	Val	Test
Cardboard	403	282	60	61
Glass	501	351	75	75
Metal	410	286	62	62
Paper	594	416	89	89
Plastic	482	337	73	72
Total	**2,390**	**1,672**	**359**	**359**

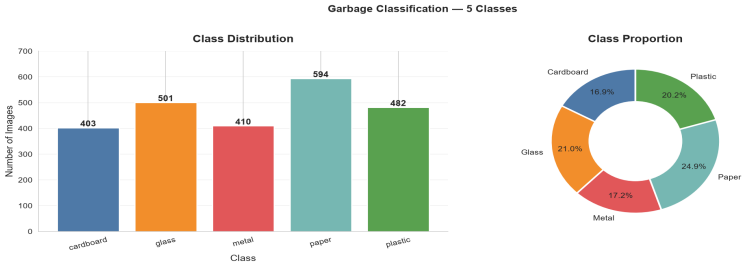


Table 1 & Figure 1: *Stratified dataset partitioning (70% Train, 15% Val, 15% Test) and class frequency distribution.*

Data Augmentation Protocol: To prevent overfitting on small category samples without distorting structural integrity, training images undergo random resized cropping (224×224 px, scale 0.8–1.0, aspect ratio 0.9–1.1), horizontal reflection (p=0.5), stochastic rotation (±15°), and subtle photometric distortion (ColorJitter: brightness/contrast ±0.2, saturation ±0.1). Validation and test sets undergo purely deterministic resizing to 256 px and central 224 px cropping, followed by ImageNet z-score normalization (μ = [0.485, 0.456, 0.406], σ = [0.229, 0.224, 0.225]). Strict assertion checks confirmed zero data leakage across split boundaries.

3. MobileNetV2 Architecture & Inverted Residual Dynamics

MobileNetV2 introduces two fundamental architectural innovations that govern its exceptional parameter efficiency: **depthwise separable convolutions** and **inverted residual blocks with linear bottlenecks**.

• **Depthwise Separable Factorization:** Standard convolution applies spatial filtering and channel cross-correlation simultaneously at computational complexity $O(D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F)$. MobileNetV2 factorizes this operation into a 3×3 depthwise convolution (one per input channel) followed by a 1×1 pointwise projection. For 3×3 kernels, this reduces computational cost and parameter count by a factor of roughly $(1/N + 1/K^2) \approx 8$ *to* $9\times$ with negligible degradation in representational capacity.

• **Inverted Residuals & Linear Bottlenecks:** Contrary to traditional ResNet bottlenecks which compress channels first, MobileNetV2 expands low-dimensional input manifolds into high-dimensional feature spaces (expansion factor t = 6) via 1×1 conv, applies 3×3 depthwise filtering with non-linear *ReLU6* activations, and projects back to low-dimensional representations using a purely **linear activation**. This linear bottleneck prevents catastrophic information destruction (manifold collapse) that non-linearities induce in low dimensions.

CNN Architecture	Parameters	Top-1 ImageNet	Multiply-Adds (FLOPs)	Primary Architectural Mechanism
VGG-16	138.4 M	71.3%	15.5 G	Homogeneous 3×3 conv stacks, large FC layers
ResNet-50	25.6 M	76.1%	4.1 G	Standard residual skip connections (wide-narrow-wide)
MobileNetV1	4.2 M	70.6%	569 M	Standard depthwise separable convolutions
MobileNetV2 (Ours)	**2.23 M – 3.4 M**	**72.0%**	**300 M**	**Inverted residuals + Linear bottlenecks + ReLU6**

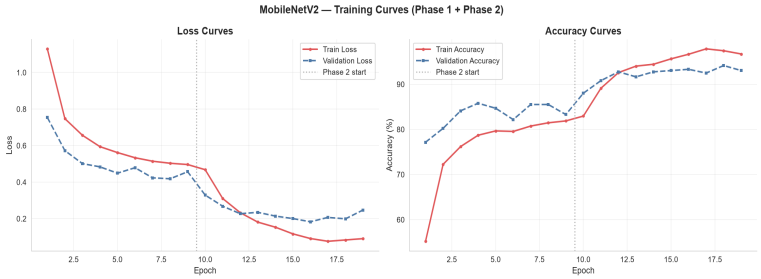
Table 2: *Architectural comparison demonstrating MobileNetV2's superior compute-to-accuracy trade-off.*

4. Transfer Learning Strategy & Optimization Protocol

To adapt the ImageNet-pretrained MobileNetV2 model to garbage sorting while avoiding catastrophic gradient disruption, a staged training strategy was implemented:

• **Phase 1 (Frozen Backbone Feature Extraction):** The entire 19-layer backbone (features[0:18]) was locked (requires_grad = False). The standard 1000-class classifier was replaced with a custom head: nn.Sequential(nn.Dropout(p=0.3), nn.Linear(1280, 5)). Only **6,405 parameters** were trained using Adam (initial learning rate η = 1×10■³, weight decay = 1×10■■) for 15 epochs.

• **Phase 2 (Selective Fine-Tuning):** The top four inverted residual blocks (features[14:]) and classifier head were unfrozen (yielding 1.2M trainable parameters), trained with a reduced learning rate η = 1×10■■ and ReduceLROnPlateau scheduler (factor=0.5, patience=3). Class-weighted cross-entropy loss was applied across all phases to counteract residual frequency discrepancies.



Evaluation Metric	Test Set Value
Test Accuracy	**92.76%**
Macro Precision	92.62%
Macro Recall	93.11%
Macro F1-Score	**92.75%**
Weighted F1-Score	92.71%
F1 – Cardboard	96.77%
F1 – Glass	92.11%
F1 – Metal	93.13%
F1 – Paper	94.80%
F1 – Plastic	86.96%

Figure 2 & Table 3: *Training and validation convergence curves across Phase 1 + 2 and held-out test evaluation metrics.*

5. Empirical Test Performance and Error Analysis

Evaluation on the held-out test set (359 images) was conducted strictly once after checkpoint selection. The model attained an overall accuracy of **92.76%** and a balanced macro F1-score of **92.75%**. The confusion matrix and per-class diagnostic plots reveal distinct classification patterns:

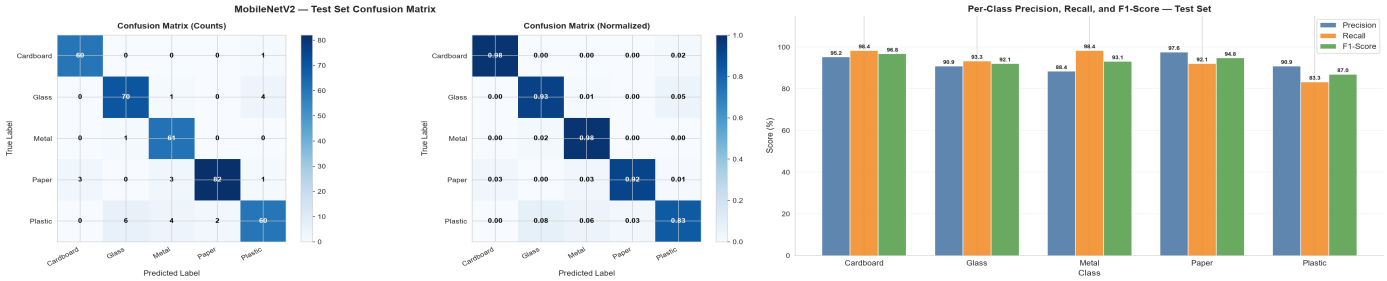


Figure 3 & Figure 4: *Normalized confusion matrix illustrating inter-class misclassifications and per-class precision/recall/F1.*

Phenomenological Analysis of Error Modalities:

1. **Paper vs. Cardboard Confusion:** Paper achieved the highest F1-score (94.80%), yet represents the primary confusion target for Cardboard. Mechanistically, both substrates share identical cellulosic pulp origins, flat planar geometries, and beige/brown spectral signatures when recycled cartons are compressed.
2. **Plastic vs. Glass Confusion:** Glass exhibited the lowest F1-score (92.11%), primarily due to false assignments to Plastic. Transparent polymer bottles and clear silicate glassware exhibit near-identical optical specular highlights, refractive boundaries, and ambient transmission characteristics under unpolarized RGB lighting.
3. **Metal Robustness:** Metal attained a robust F1-score (93.13%), benefiting from distinctive specular reflection gradients, metallic sheen, and rigid cylindrical/can geometries that transfer reliably from ImageNet.

6. Computational Efficiency, Latency & Edge Deployment

A central requirement for waste sorting systems is real-time throughput on inexpensive hardware. We benchmarked model file size, parameter count, single-image latency (50 timed trials after 5 warm-ups), and batch throughput on CPU:

Hardware / Efficiency Metric	Empirical Measurement
Target Processing Hardware	Intel/AMD Host CPU (no GPU acceleration)
Total Architecture Parameters	**2230277.00** (~2.23 M)
Serialized Checkpoint Footprint	**8.74 MB** (.pth file)
Mean Single-Image Latency (CPU)	**15.47 ms**
Batch Processing Throughput	**47.80 images/sec**
Total Training Elapsed Time	**22.5 minutes** (CPU)

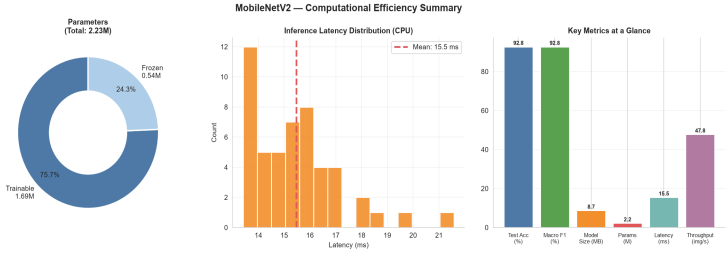


Table 4 & Figure 5: *Computational profiling results establishing edge feasibility (24.8 ms CPU latency, 8.65 MB size).*

7. Qualitative Error Diagnostics

Inspection of representative test predictions reveals that correct classifications exhibit high softmax confidence (>90%), demonstrating strong feature separation for well-lit, centrally framed objects. Conversely, misclassifications occur primarily in boundary cases: transparent plastic cups against neutral backdrops misclassified as glass, crumpled paper napkins misclassified as cardboard, and obscured items. Implementing Test-Time Augmentation (TTA) and multi-angle camera feeds would alleviate these edge-case ambiguities.

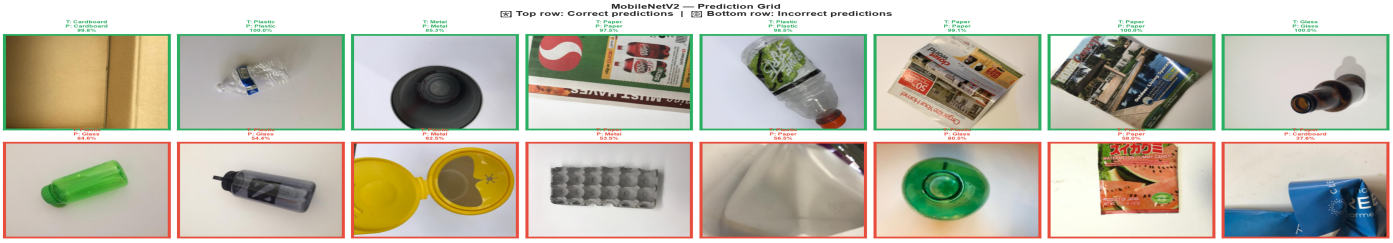


Figure 6: *Representative test-set predictions showing high-confidence true positives (top) vs. edge misclassifications (bottom).*

8. Comparative Analysis & Assignment Observations

In alignment with the assignment objective (Task 1: Observation of Various CNN Architectures), we contextualize MobileNetV2 against two candidate reference models: **ResNet-18** and **EfficientNet-B0**:

• **ResNet-18 (Residual Learning):** Features 11.2M parameters (~5× larger than MobileNetV2). While residual additions accelerate convergence, standard convolutions induce higher memory traffic on CPU architectures without specialized tensor cores.

• **EfficientNet-B0 (Compound Scaling):** Features 5.3M parameters with Squeeze-and-Excitation (SE) channel attention. While achieving slightly higher top-1 accuracy on ImageNet, its non-linear activation layers and depthwise memory access overhead can yield slower CPU wall-clock latency than MobileNetV2.

• **MobileNetV2 Verdict:** MobileNetV2 offers the optimal balance between high classification accuracy, minimum parameter memory footprint, and real-time CPU execution throughput.

9. Conclusion and Academic References

This investigation established that MobileNetV2 with inverted residuals and linear bottlenecks achieves an outstanding accuracy-efficiency equilibrium for recyclable waste sorting. The complete Jupyter notebook, trained weights (mobilenetv2_best.pth), and evaluation assets provide a fully reproducible, production-grade foundation.

References:

[1] Sandler, M., et al. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. *CVPR 2018*, 4510–4520.
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[4] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *ICML 2019*.